Structural results for La Palma island using 3-D gravity inversion

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[1] A recent gravity survey composed of 317 bench marks all over the island of La Palma (Canary Islands) is used, in combination with satellite data for regional aspects, to obtain results about structural properties of the island connected with the tectonic environment and local volcanism. To that end, a nonlinear three-dimensional gravity inversion approach is considered. The inversion scheme provides, in a nonsubjective form, the geometry of the anomalous bodies constructed in a random growth process. Results from the inversion can be interpreted in the framework of the geologic evolution of this ocean island volcano as a complex composite volcano with a large central body with high-density corresponding to the older intrusive part of the basalt complex. New unexpected features are enlightened, such as large thermal anomalies in the upper mantle southward of La Palma, as well as fracture en echelon zones associable to a slow active process of dislocation related to the recent volcanism in the southern half of the island. The results obtained for La Palma as a test site testify to the usefulness of the developed gravity inversion methodology for structural studies on islands in general.

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1. Introduction: Geological Setting

[2] Rift zones on ocean island volcanoes are elongated areas or morphological ridges along which volcanic and intrusive activity are concentrated. Being the locus of major volcano growth, they have strong control over magmatic processes, rates, and pathways of magma transport, and structure and stability of volcanic edifices [*Galipp et al.*, 2006].

[3] According to *Walker* [1999], rift zones and underlying intrusion swarms occur in most volcanoes and contain the paths taken by magmas moving through the crust. Most of these intrusive complexes are positioned and oriented by tectonic structures or may be propagated laterally from volcanic centers along rift zones, following neutral buoyancy levels. Closer to the surface, swarms of sheet intrusions (dikes and sills) represent conduits for magma transport from the reservoirs to the surface or to shallow intrusions, the setting of this latter type of intrusions being well documented in active volcanoes, such as Kilauea or Piton de la Fournaise [*Malengreau et al.*, 1999]. *Walker* [1999] also suggests that insertion of dikes could cause structural changes such as bending or initiation of a rift zone.

[4] According to *Rymer and Brown* [1986], positive gravity anomalies characterize mainly basaltic volcanoes, and are caused by a relatively dense intrusive complex/magma body, which contrasts with its surroundings either because that intrusive body is more mafic than average or, more likely, because near surface previously erupted materials are uncompacted with a higher degree of vesiculation. For Hawaiian volcanoes, *Walker* [1992] considers a complex with density 2800 kg m^{-3} juxtaposed against highly vesicular lava flows having a much lower density of about 2000 kg m^{-3}. *Ryan* [1987] proposed that complexes of hypovolcanic intrusions and cumulates develop upward during growth of a volcano. As a result, the center of a mature oceanic shield volcano is expected to be characterized by a column-like body of high-density rock. Apart from the general ambiguity problem of the gravity inversion, some problems arise from the accessibility and sharp topographic effects in volcanic areas mainly formed by very high bulk density material. Therefore information about intrusive complexes could be obtained by gravity study and further inversion of the observed anomaly [*Camacho et al.*, 2007].

[5] La Palma is the youngest and historically most active island of the Canarian Archipelago (Figure 1), a 500 km long chain of seven intraplate volcanic islands off the NW Africa coast. Subaerial volcanic activity at the archipelago started more than 20 Ma ago with the formation of the eastern islands, Lanzarote and Fuerteventura [*Carracedo et al.*, 1998]. Tenerife, La Palma, and El Hierro are presently in the shield stage of growth, whereas Lanzarote, Fuerteventura, Gran Canaria, and La Gomera are in the posterosional stage of evolution [*Carracedo*, 1999].

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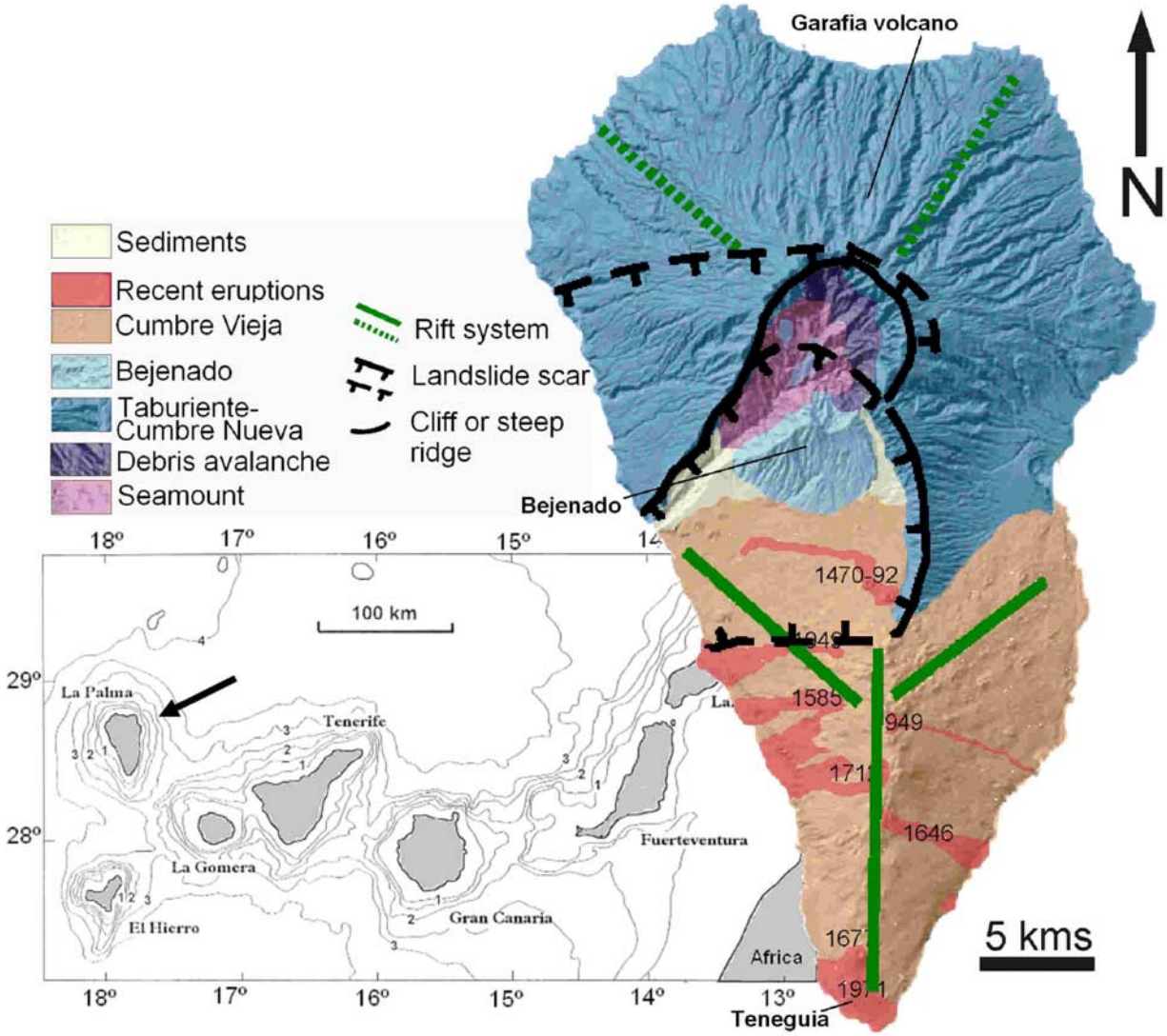

Figure 1. Geographical location (latitude and longitude) and geological elements (modified from *Urgeles et al.* [1999] and *Carracedo et al.* [1999]) of La Palma Island. Some location names are also included.

[6] Although some typical hot spot features such as long-wavelength gravity high or a midplate topographic bulge are absent [*Watts*, 1994], seismic reflection studies suggest the existence of a moderate bathymetric anomaly in the westernmost islands of La Palma and El Hierro [*Ranero et al.*, 1995; *Urgeles et al.*, 1999]. Bathymetric elevation could reflect a thermal anomaly in an upper mantle residue from an old plume, and therefore it does not carry (or does so in a highly diluted form) the typical geophysical and geochemical plume signatures; the fractures are well developed on the continental and oceanic crust, but not in the extremely thick sedimentary pile between the Canary Islands and Africa; and the Canary Islands uplift took place through transpressive shears, and not by means of purely reverse faults [*Anguita and Hernán*, 2000].

[7] La Palma consists of an old radial shield volcano (Taburiente) that began to form a dominant rift at about 800 ka, which ultimately resulted in a presently active, well-developed rift zone (Cumbre Vieja). The geology of the La Palma island is summarized by *Carracedo et al.* [1999, 2001] and *Galipp et al.* [2006] and can be divided into three major units (Figure 1): (1) the old basaltic complex (circa 4.0 to 3.0 Ma) which comprises a Pliocene seamount sequence and a plutonic complex, uplifted and tilted by intrusions coeval with the later subaerial activity; (2) the old volcanic series (1.7 to 0.5 Ma) which includes the Garafia volcano, the Taburiente shield volcano, the Bejenado edifice, and the Cumbre Nueva series; and (3) the Cumbre Vieja series (125 ka to present) which is confined to the southern half of the island.

[8] The volcanic evolution of the island encompasses two different periods: a first stage of submarine growth, followed by the effusion of subaerial volcanic rocks which are composed of several successive shield volcanoes. Submarine stage of growth is almost entirely recorded by basaltic series and plutonic rocks. These materials crop out in the

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northern half of the island, in the inner part of the Caldera de Taburiente [*Fernández et al.*, 2002]. Subaerial stage is characterized by several episodes of gravitational collapse alternate with the building of volcanic edifices [*Fernández et al.*, 2002]. Recent volcanic activity is confined to the southern part of La Palma (Cumbre Vieja) and located along a north–south trending ridge, which constitutes a volcanic rift zone (in the sense of *Walker* [1992]). Recent volcanics include more than 120 cinder cones [*White and Schmincke*, 1999]. At least seven eruptions took place in this area during the last 500 years.

[9] A large area of debris avalanche deposits has been discovered on the western submarine flanks of La Palma [*Urgeles et al.*, 1999]. The island may have had two giant landslides, although the existence of the older one has not yet been unambiguously verified. Both affect the northern, Plio-Quaternary Taburiente shield volcano. The last one, the Cumbre Nueva giant landslide, took place at about 560 ka and involved a volume possibly exceeding 200 km^3 [*Carracedo et al.*, 1999]. The most recent eruption near the summit of the Cumbre Vieja, that of 1949, was accompanied by development of a west facing normal fault system along the crest of the volcano. The geometry of this fault system and the timing of its formation in relation to episodes of vent opening during the eruption indicate that it is not the surface expression of a dyke. Instead, it is interpreted as being the first surface rupture along a developing zone of deformation and seaward movement within the western flank of the Cumbre Vieja: the volcano is therefore considered to be at an incipient stage of flank instability [*Day et al.*, 1999]. *Ward and Day* [2001] have suggested the possibility that during a future eruption, Cumbre Vieja Volcano on the Island of La Palma may experience a catastrophic failure of its west flank, dropping 150 to 500 km^3 of rock into the sea. They modeled the subsequent tsunami corresponding to such an estimated collapse, concluding that the waves generated by the event could transit the entire Atlantic, reaching America with a height of about 10 m. This suggestion has renewed the interest and controversy about the structure and volcano-tectonic evolution of the island.

[10] To identify any displacement of the western flank overlying the discontinuity, a ground deformation network was established on the volcano. The results obtained by *Moss et al.* [1999] were within the error margins of the techniques employed, but they suggest a possible coherent westward displacement of stations to the west of the 1949 fault system.

[11] Geophysical data provide a valuable approach to gain a more complete picture of volcanic edifices and subsurface structures, including the presence of magma reservoirs [e.g., *Rousset et al.*, 1989]. In the western Canary Islands, several types of geophysical studies have been carried out. Sonar images and seismic reflection data has been used to identify major landslide events [*Urgeles et al.*, 1999]; marine magnetic measurements to locate fractures and magmatic underplating [*Catalán et al.*, 2003]; bathymetry and marine gravity to identify a sedimentary cover masking the oceanic swell [*Canales and Dañobeitia*, 1998]; local terrestrial gravity to detect crustal structures [*Gottsmann et al.*, 2008; *Montesinos et al.*, 2006]; seismic refraction and gravity to determine the crustal structure [*Bosshard and MacFarlane*, 1970]; multichannel seismic reflection to study the lithospheric response to volcanic loading [*Collier and Watts*, 2001]; magnetic surveys to model the feeding system [*Blanco-Montenegro et al.*, 2008]; gravity, magnetic, and seismic refraction and reflection profiles to constrain the crustal structure as well as the reflector geometry [*Collier et al.*, 1998]; and high-quality seismic data to investigate the anelastic attenuation of seismic energy [*Canas et al.*, 1995].

[12] Gravity studies in volcanic areas in particular can provide unique insight into shallow subsurface density anomalies associated with the structural and magmatic history of an active volcanic system [*Rymer and Brown*, 1986]. In this paper, we analyze new gravity data recently observed inland to obtain, via nonsubjective inversion, some structural conclusions about the subsurface mass properties corresponding to the volume below La Palma.

2. Gravity and Elevation Data

[13] The data used for this structural study are mainly composed of gravity and elevation values for La Palma. For terrestrial areas, a gravity survey was carried out in 2006 covering the island with 318 bench marks. The corresponding elevation values were determined simultaneously by GPS observation. As complementary data, we used a digital elevation model for the island composed of 129,000 terrestrial points gridded with a step of 25 m provided commercially by the Instituto Geográfico Nacional [*Centro Nacional de Información Geográfica*, 2000].

[14] For the surrounding marine areas, a set of satellite altimetry data was considered to provide information about gravity anomaly, geoid and bathymetry.

2.1. Survey Gravity Data

[15] The terrestrial gravity survey was carried out in June and November 2006 with a LaCoste and Romberg model G gravimeter equipped with a digital recording device. The total number of observations was 385, corresponding to 318 bench marks (317 in La Palma and 1 reference station in Tenerife) (Figure 2). Bench marks were distributed as homogeneously as possible taking into account the very high roughness of the island, with an average mutual distance between neighboring stations of 1400 m (Figure 2). The total number of bench marks with repeated observations is 28 and the number of redundant observations is 385–318 = 67, which means about 17% of the total.

[16] Simultaneously, elevation values for the bench marks were determined by means of GPS differential observations with an estimated accuracy of the resulting values mostly better than 10 cm.

[17] Tidal corrections according to the Cartwright-Tayler development (484 waves) were considered. Then, we carried out a traverse adjustment of the gravity data for the repeated observations. As main results we obtained the instrumental drifts (-0.113 ± 0.004 mGal d^{-1} in June and -0.168 ± 0.011 mGal d^{-1} in November), the root mean square residual of observations (0.071 mGal) and an estimation of the internal accuracy of the adjusted gravity data (0.068 mGal) (1 mGal = 10^{-5} m s^{-2}). The resulting relative gravity values were referred to a previous value (979,291 mGal) observed in the neighboring island of Tenerife and connected with the absolute reference IGSN71.

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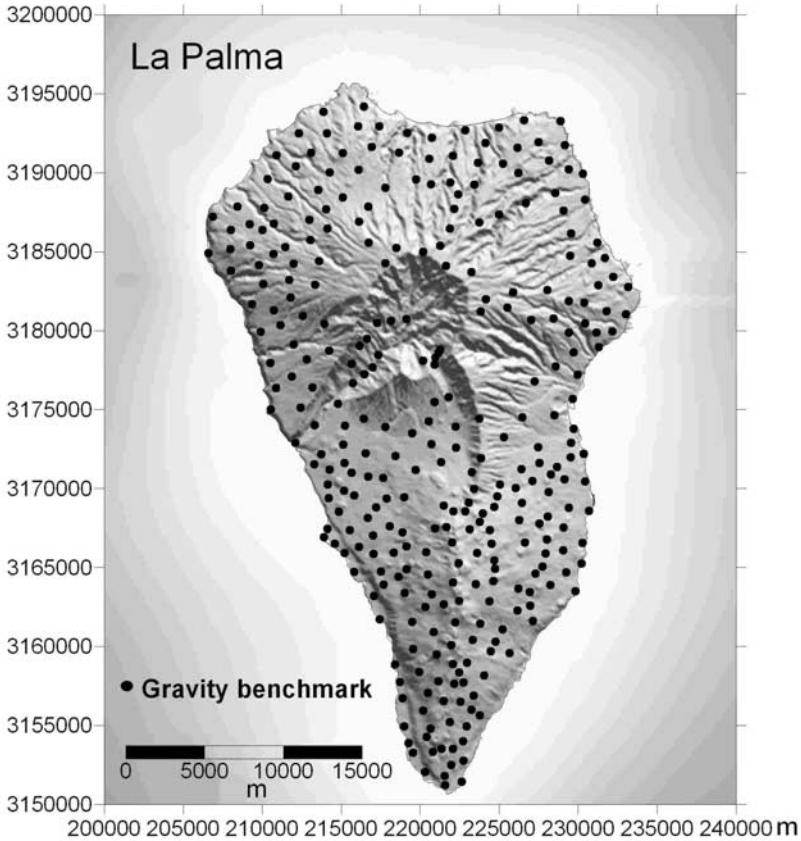

Figure 2. Digital elevation model of La Palma Island and location of the gravimetric bench marks (318 points). UTM (m) coordinates.

[18] By combining the gravity and GPS elevation data, we calculated the gravity disturbances (the precise term "gravity disturbances" is employed instead "gravity anomalies" taking account that we use GPS ellipsoidal heights instead of geoidal heights). First, computation of the normal gravity was referenced to GRS80 [*Moritz*, 1980]. Then, a free-air gradient of -0.308 mGal m^{-1} and a value for average density of 2300 kg m^{-3} were considered to be applied for the Bouguer correction. This last value for average density was determined from the gravity data by looking for a minimum correlation between elevation and gravity disturbances for the shortest wavelength components of topography. The further inversion process confirms to us that this value is suitable.

[19] Correction for topographical irregularities is accomplished by using a digital elevation model (DEM) composed by 129,000 points (Figure 2). This model is composed by a grid of points, 25 m step, for the terrestrial area plus a satellite bathymetry up to a distance of 100 km with a 3 km step obtained from http://topex.ucsd.edu/cgi-bin/get_data.cgi [*Sandwell and Smith*, 1997] (Figure 3). To avoid problems of coherence, the height of the bench mark in DEM is used here instead of the observed GPS value. Very large correction values for the topographical effects were obtained. They range from 12.5 mGal to 62.4 mGal, with an average value of 26.2 mGal.

[20] By including the terrain correction for the gravity disturbances we computed a Bouguer disturbances (with terrain correction) map (see Figure 4). Values for Bouguer disturbances are distributed around an average value of 247.17 mGal with a standard deviation of 25.50 mGal. They show a clear high value that dominates most of the island. The southern extreme, where the most recent activity has been recorded, is characterized by the lower disturbance values.

2.2. Satellite Data

[21] According to *Bosshard and MacFarlane* [1970], a clear gravity regional trend occurs in the Canary Islands. It is associated to the crust thickening toward the African continent. Taking into account that the terrestrial gravity values for La Palma are heavily perturbed by sharp local density anomalies, a suitable determination of the gravity regional trend must be obtained by using data from the surrounding oceanic areas. Accordingly, we loaded the geoid, bathymetry and free-air values (deduced from satellite altimetry for oceanic areas) contained in the ASCII gravity anomaly and geoid height files provided at http://topex.ucsd.edu/marine_grav/mar_grav.html [*Sandwell and Smith*, 1997] from Geosat and ERS-1 satellite altimetry. We considered the surrounding area to a distance of 100 km, with a step of about 3 km as provided in the files.

[22] Starting from the free-air anomaly given by the satellite data and using data for bathymetry and geoid height, we determined (taking 2600 kg m^{-3} as mean density for the crustal density) values for Bouguer disturbances (converting

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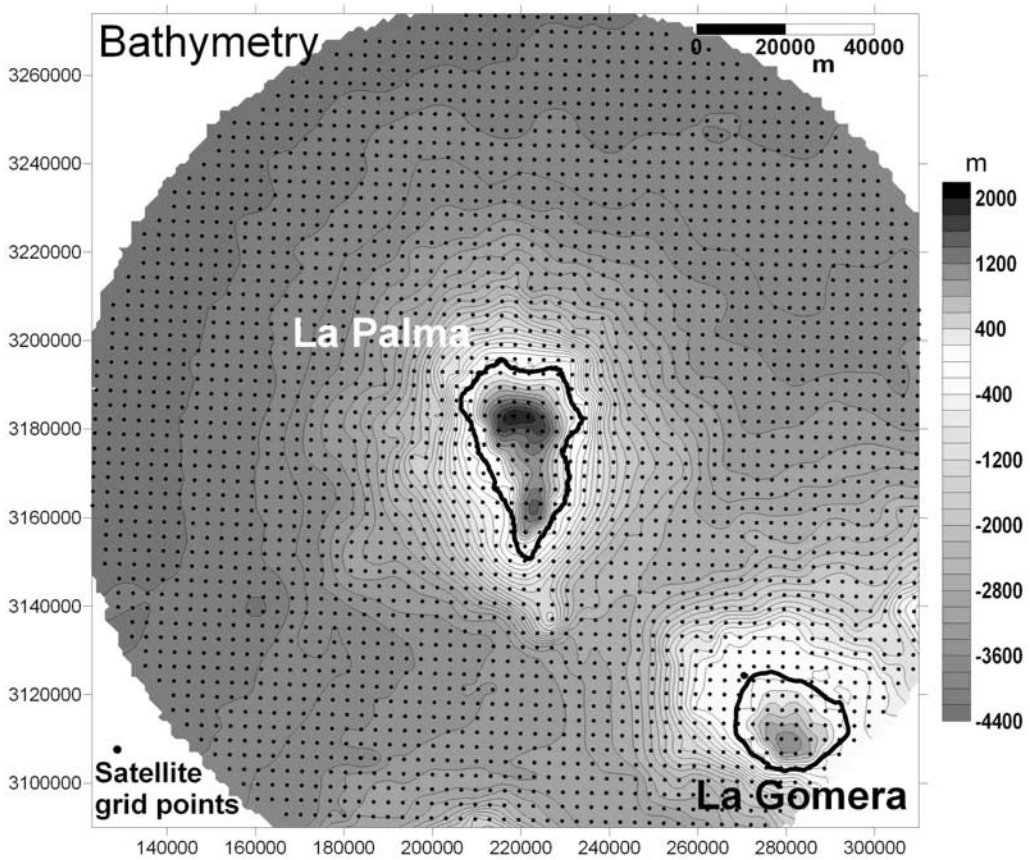

Figure 3. Bathymetry from altimetric satellite data. Data grid points shown. UTM coordinates.

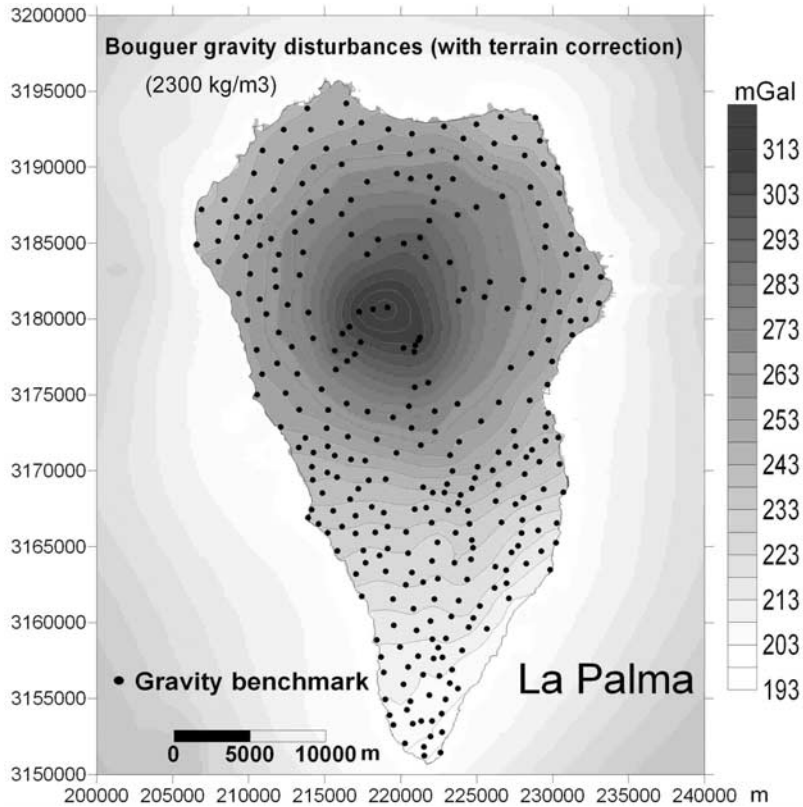

Figure 4. Refined Bouguer disturbances also including the terrain correction for mean density 2300 kg m^{-3}. Positions of bench marks are also included. UTM coordinates are used for axes.

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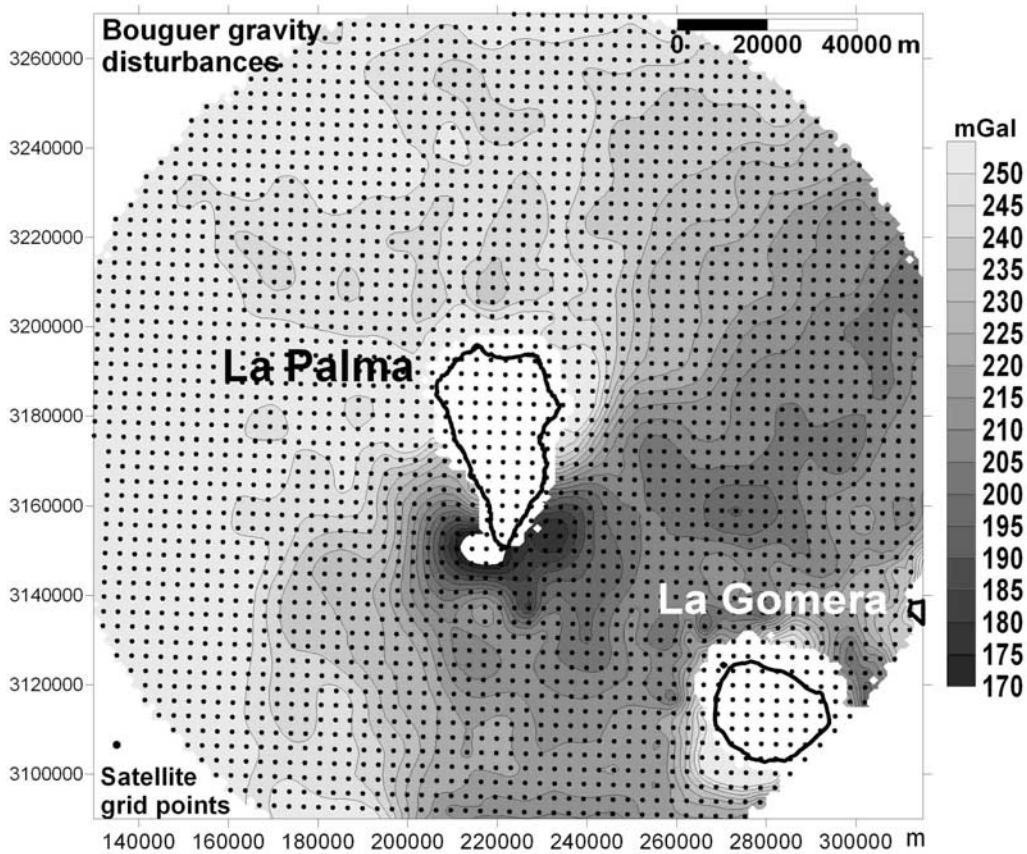

Figure 5. Regional Bouguer disturbances obtained by combining free-air anomaly, bathymetry, and geoid height from satellite data in the area. Assumed mean crust density is 2600 kg m^{-3}. UTM coordinates are used. A linear trend of about 0.46 (±0.01) mGal km^{-1} with SE–NW strike of azimuth 122°NE is estimated.

anomalies previously to disturbances by using the geoid heights). Figure 5 shows a view of this regional Bouguer disturbance. It will correspond mostly to the thickness pattern of the crust in this area. A noticeable low is observed close to the SE border of La Palma. This could be due to the presence of a thermal anomaly corresponding to recent volcanism. The last historical eruption in the Canary Island took place in this area in 1971 (Teneguia volcano).

3. Gravity Inversion Method

[23] The inverse gravimetric problem, namely the determination of a subsurface mass density distribution corresponding to an observed gravity anomaly or some observed gravity disturbances, has an intrinsic nonuniqueness in its solution [e.g., *Al-Chalabi*, 1971]. Moreover, data must be considered as insufficient (discrete number of points) and inaccurate (instrumental limitations). Nevertheless, particular solutions can be obtained by including additional constraints about the model parameters (subsurface structure) and about the data parameters (statistical properties of the inexact data, e.g., Gaussian distribution). The inversion methods that define the geometrical properties of anomalous bodies with prescribed density contrast [e.g., *Pedersen*, 1979; *Barbosa et al.*, 1997] correspond to a nonlinear methodology and offer interesting results, limited to the validity of the hypothesis used. For the fully nonlinear treatment, the methods of exploration of the space model often give the best option [*Tarantola*, 1988]. This exploration process can be conducted randomly [*Silva and Hohmann*, 1983].

[24] In previous papers [*Camacho et al.*, 2000, 2002, 2007], an original methodology for gravity inversion has been described and applied. It looks for an anomalous model defined as 3-D aggregation of M parallelepiped cells which are filled, in a growth process, by means of prescribed positive and/or negative density contrasts. The design equations corresponding to gravity anomaly values Δg_i (also for gravity disturbances) observed in N bench marks (x_i, y_i, z_i) are

$$\Delta g_i = \sum_{j\in J^+} A_{ij}\Delta\rho_j^+ + \sum_{j\in J^-} A_{ij}\Delta\rho_j^- + \delta g_{reg} + \delta g_{top} + v_i, \quad i = 1, \ldots, N, \tag{1}$$

where A_{ij} is the vertical attraction for unit density for the jth parallelepiped cell upon the ith observation point, $\Delta\rho_j^-$, $\Delta\rho_j^+$ are prescribed density contrasts (negative and positive) for the jth cell, J^+, J^- are sets of indexes corresponding to the cells filled with positive or negative density values, and δg_{reg},

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δg_{top} are optional terms for additional values corresponding to regional trend and topographical correction given by

$$\delta g_{reg} = p_0 + p_x(x_i - x_M) + p_y(y_i - y_M), \qquad i = 1, \ldots, N, \quad (2)$$

$$\delta g_{top} = \delta\rho_T C_i.$$

Here p_0, p_x, p_y are parameters for the linear trend, x_M, y_M are averaged coordinates, C_i are coefficients for the terrain correction, and $\delta\rho_T$ is an additional value with respect to the initially adopted terrain density. In these equations, J^+, J^- (cells filled with positive and negative density contrast), p_0, p_x, p_y, and $\delta\rho_T$ (regional parameters) are the main unknowns to be determined.

[25] The problem of nonuniqueness is solved by means of assuming a mixed fit condition about minimization of the l_2 fitness and the smoothness of the model:

$$\boldsymbol{v}^T \boldsymbol{Q}_{\boldsymbol{D}}^{-1} \boldsymbol{v} + \lambda\, \boldsymbol{m}^T \boldsymbol{Q}_{\boldsymbol{M}}^{-1} \boldsymbol{m} = \min., \quad (3)$$

where $\boldsymbol{m} = (\Delta\rho_{1_1}, \ldots, \Delta\rho_M)^T$ are density contrast values for the M cells of the model, $\boldsymbol{v} = (v_1, \ldots, v_N)^T$ are residual values for the N data points, $\boldsymbol{Q_D}$ is an a priori covariance matrix for the gravity data, $\boldsymbol{Q_M}$ is an a priori covariance matrix for the cell aggregation geometry, and λ is factor for selected balance fitness/smoothness of the model. For practical applications λ is selected to produce uncorrelated inversion residues.

[26] The operational method for solving the design system subjected to the fit condition is a controlled 3-D growth (everywhere) of the anomalous bodies by means of exploratory approach. For the kth step of the growth process, k cells have been filled with the prescribed density contrast giving rise to a calculated gravity value

$$\Delta g_i^c = \sum_{J_k^+} A_{ij}\Delta\rho_j^+ + \sum_{J_k^-} A_{ij}\Delta\rho_j^-, \quad J_k^+ + J_k^- = k \quad (4)$$

Then for the $(k + 1)$th step, we try to fill a new cell by trying the prescribed density contrasts on every j empty cell according to the fit conditions:

$$\Delta g_i - \left(\Delta g_i^c + A_{ij}\Delta\rho_j\right) f - \left(p_0 + p_x(x_i - x_M) + p_y(y_i - y_M)\right) - \rho_T C_i = v_i \qquad i = 1, \ldots, n, \quad (5)$$

$$\boldsymbol{v}^T \boldsymbol{Q}_{\mathrm{D}}^{-1} \boldsymbol{v} + \lambda f^2\, \boldsymbol{m}^T\, \boldsymbol{Q}_{\boldsymbol{M}}^{-1} \boldsymbol{m} = \min, \quad (6)$$

where $f > 1$ is a scale factor to allow for a fit between the anomaly of the provisional (not totally developed) model and the observed anomaly. Once f, p_0, p_x, p_y and $\delta\rho_T$ are determined by linear fit, we consider a characteristic value for the test jth cell given by

$$e_j^2 = \boldsymbol{v}^T \boldsymbol{Q}_{\mathrm{D}}^{-1} \boldsymbol{v} + \lambda f^2\, \boldsymbol{m}^T\, \boldsymbol{Q}_{\mathrm{M}}^{-1} \boldsymbol{m} \quad (7)$$

We choose as optimal cell for the $(k + 1)$th step that j cell giving

$$e_j^2 = \min. \quad (8)$$

The process continues until we reach $f \cong 1$. As a final result, we arrive nearly automatically at a 3-D model described as the aggregation of cells filled with the prescribed density contrast. It fits the observed values well enough and keeps a small model anomalous mass involved.

[27] *Camacho et al.* [2000, 2002] and *Gottsmann et al.* [2008] give some simulation examples showing the suitability of this 3-D inversion approach and also pointing out the possible distortions and limitations. Results are quite satisfactory. A certain trend to produce rounded bodies or other distortions is observed for the low-reliability zones. This inversion approach has been applied to the local gravity data of La Palma.

4. Inversion Model for La Palma

4.1. Regional Trend

[28] The surrounding Bouguer disturbances (given in Figure 5 and obtained from satellite data) can be used to estimate a smooth regional trend for the local gravity disturbances of La Palma. By using the same general inversion approach described and applied below for the data of the island, we determined a linear trend of about 0.46 (±0.01) mGal km^{-1} with a SE–NW strike of azimuth 122° NE. This linear trend will represent the very long wavelength component of the local disturbances. A general average value of 0.2 mGal km^{-1} was obtained by *Bosshard and MacFarlane* [1970] from west of La Palma–Hierro to Gran Ganaria and associated to the crust thickening toward the African continent.

4.2. Local Gravity Disturbances

[29] By subtracting this linear trend from the inland gravity disturbances we calculated a local disturbance (see Figure 6), which corresponds mostly to local density structures and will be considered for the further inversion process.

[30] Again, the obtained gravity disturbances are characterized by a large maximum centered in the Caldera de Taburiente (shield volcanism), and that affects nearly the whole island. The southern edge, corresponding to the last volcanism in the island, is affected by a local minimum. The total range of variation of the local anomaly is around 125 mGal. Our intention is to construct a model for distribution of anomalous bulk density below the surface of the island, capable of producing the observed gravity disturbances.

[31] The local Bouguer disturbance Δg_i values for N = 316 gravity stations are considered for an automatical gravity inversion according to the previously described methodology. First, a partition of the subsoil volume of the island into 25,000 parallelepiped cells is considered. The cells have sides ranging from 282 m in the shallowest zone (1800 m above sea level) to 1700 m for depth 22,000 m below sea level (bsl), with an average side of 500 m. Matrix $\boldsymbol{Q_D}$ for a priori covariance of data is assumed to be: $\boldsymbol{Q_D} = \boldsymbol{I_N}$ 4.9, where $\boldsymbol{I_N}$ is a unit matrix and a value 0.7 mGal is assumed as standard deviation for the gravity data. Matrix $\boldsymbol{Q_M}$ for a priori covariance of the cell distribution geometry can be

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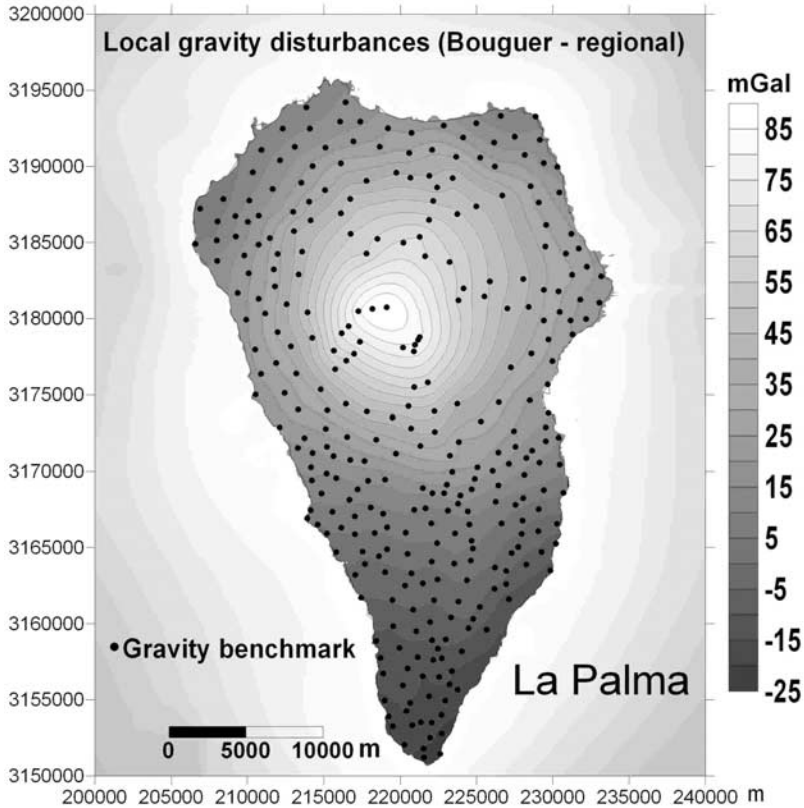

Figure 6. Local gravity obtained from the refined Bouguer disturbances by subtracting the regional trend. This gravity disturbances are further employed for the inversion process in order to determine the local structure under the island. UTM coordinates are used.

taken as a diagonal normalizing matrix of nonnull elements that are the same as the diagonal elements of $\boldsymbol{A}^T\boldsymbol{Q}_{\boldsymbol{D}}^{-1}\boldsymbol{A}$, where $\boldsymbol{A}$ is the design matrix with elements A_{ij}. The only remaining question to decide is to choose the values of the prescribed values (positive and negative) that will be used as extreme values to fill the cells of the model. The value λ is automatically determined by assuming the condition of null autocovariance for the final residuals.

[32] We fit a model of anomalous structure neglecting the possible properties of the medium. To that end, we assume everywhere extreme density contrasts of −350 and 420 kg m^{-3}. These values have been selected by trying several possibilities and choosing those values that give rise to balanced shaped models for the anomalous bodies (too low contrasts produce connected, convex and rounded volumes; too high contrast produce disaggregated and "skeleton" volumes). By including a smoothing effect for the density contrast in the boundary of the anomalous bodies [*Camacho et al.*, 2002] from the former extreme values, the average anomalous density values for the model are −306 and 351 kg m^{-3}. A detailed description of the final inversion model is shown in Figures 7 and 8 by means of some horizontal cross sections for several depths and some vertical profiles for W–E and S–N sections. The final inversion residuals have a mean square root value of 0.57 mGal, which is assumed as indicative of the inner coherence of the gravity data and will correspond to very local gravity disturbances and to local inaccuracies of the topographical gravity correction.

[33] Several interesting morphological features can be observed in the adjusted model. The most relevant structure is clearly the high-density body under the northern half of the island (body A in Figure 9). This truncated vertical prism-shaped body contains a total anomalous positive mass of 1.01×10^{15} kg. The horizontal sections of this body (a and b in Figure 7) have an average diameter of about 14 km, and vertically (c and d in Figure 7) it extends from near the terrain surface (with a possible outcropping in the bottom of the Caldera de Taburiente depression) to a depth of about 14 km bsl (see vertical profiles in Figure 8). In Figure 9 we can see some interesting aspects of this body. At a depth of about 6 km bsl (Figure 9c) it is quite regularly shaped, with a particularly straight edge at its SW border (with azimuth about N110°E) and also at its SE border. It suggests some ancient tectonic alignments. Some offshore predicted lows in the SW and SE borders of Figure 9c favor the assumption of a regional effect. It should be noted that the southern portion of body A is more centered with respect to the gravity survey and therefore there is a higher confidence with respect to the northern portion. The cross section for 2000 m bsl also shows a noticeable feature: the apparent elongation of the central high southward (azimuth N130°E). This elongation is not revealed at deeper or shallower depth levels.

[34] The southern half part of the model shows clearly different features. There are no high-density bodies, and the most relevant structure is an N–S alignment (azimuth of

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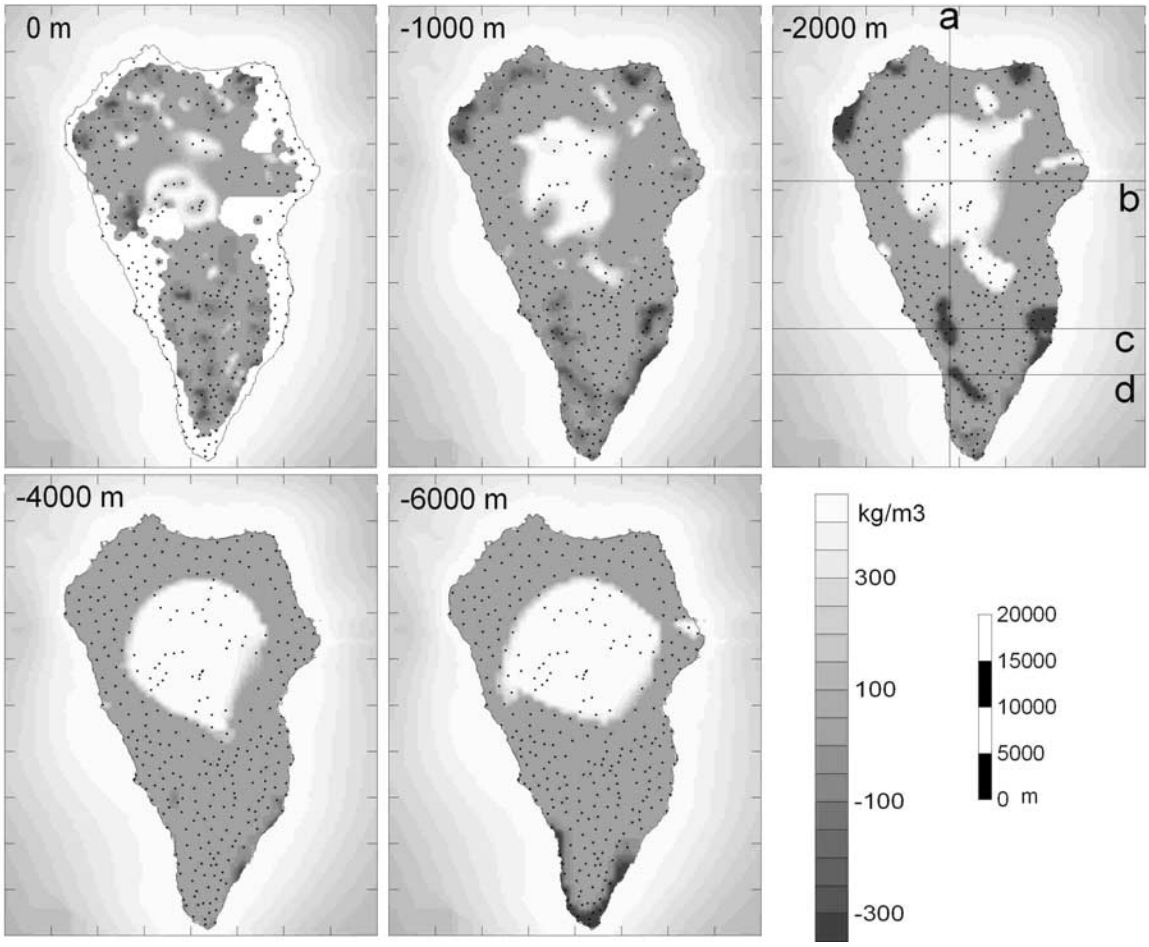

Figure 7. Horizontal cross sections of the 3-D model of anomalous density for depths 0, −1000, −2000, −4000, and −6000 (negative bsl). Section for 2000 m bsl includes the cross lines (a,b,c,d) for the vertical profiles (Figure 8).

N175°E) of low-density bodies located in the western slope with a mean depth of about 2 km bsl This low-density structure involves a total anomalous mass defect of about 1.2×10^{13} kg, and it extends a total length of 14 km. For the section corresponding to depth 2000 m bsl (Figure 9b), we observe some interesting low-density structures, mainly on the western flank of the ridge, formed by several small en echelon tracks with directions not far from that of the SW side of the main body. For vertical profiles, Figure 8 also shows (profiles in Figures 8a, 8c, and 8d) the location of these low-density small bodies in the southern half of the island.

[35] Two other low-density structures are located in this southern half of the island: one located in the eastern side with a depth similar to the main low body (about 3 km bsl) and another, very deep one (about 10 km bsl), which is predicted southward out of the island. At a very shallow level (Figure 9a), we observe an alignment of low-density small bodies in the summit line, corresponding to the alignment of the last eruptive activity on the island.

[36] Some other low-density bodies are located, at shallow levels, in the northern tip of the island, around the main positive body. They are located mostly in peripheral sites with respect to the survey and play the role of defining the structural limits. Further observational work would be required to define them better.

5. Discussion and Conclusions

[37] According to the ideas of *Walker* [1999] and *Rymer and Brown* [1986], given in section 1, the center of a mature oceanic shield volcano is characterized by a column-like body of high-density rocks. The main anomalous structure detected in La Palma is the central extended density maximum (body A in Figure 9), which corresponds to the intrusive and plutonic part of the basal complex outcropping under Caldera de Taburiente. It can be interpreted as the Pliocene age uplifted seamount and a relatively dense intrusive plutonic complex/magma body, which contrasts with its surroundings mostly because near the surface, erupted materials are uncompacted, with either a higher degree of vesiculation, cracks or pores. This extended body constitutes the core of the successive overlapping northern shield volcanoes (Seamount-Taburiente-Cumbre Nueva volcanoes). Its corresponding lithologies and structural features have got repercussion at shallower levels. Relative high-

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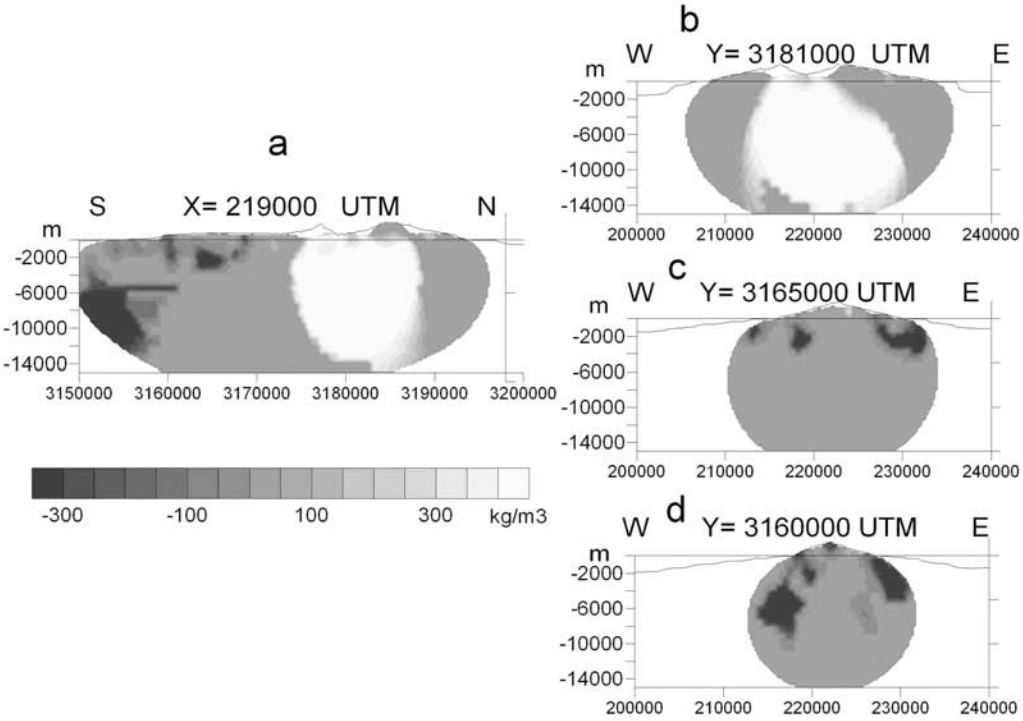

Figure 8. Vertical profiles of the 3-D inversion model for anomalous density. The corresponding cross lines are indicated in Figure 7. UTM coordinates are included. Standard deviation of residuals is 0.57 mGal.

and low-density subsurface volumes may reflect in some way the rock storage properties. So, for instance, the upper boundary of this high-density body could represent an impermeable boundary (basement) for the groundwater storage, so contouring groundwater table levels [*Consejo Insular de Aguas de La Palma*, 2001] follow the limits of the high-density body (Figure 10).

[38] The elongation of body A southward, pointed out in Figure 9, suggests a possible change of the magmatic pattern toward the volcanism of Cumbre Vieja complex. *Galipp et al.*

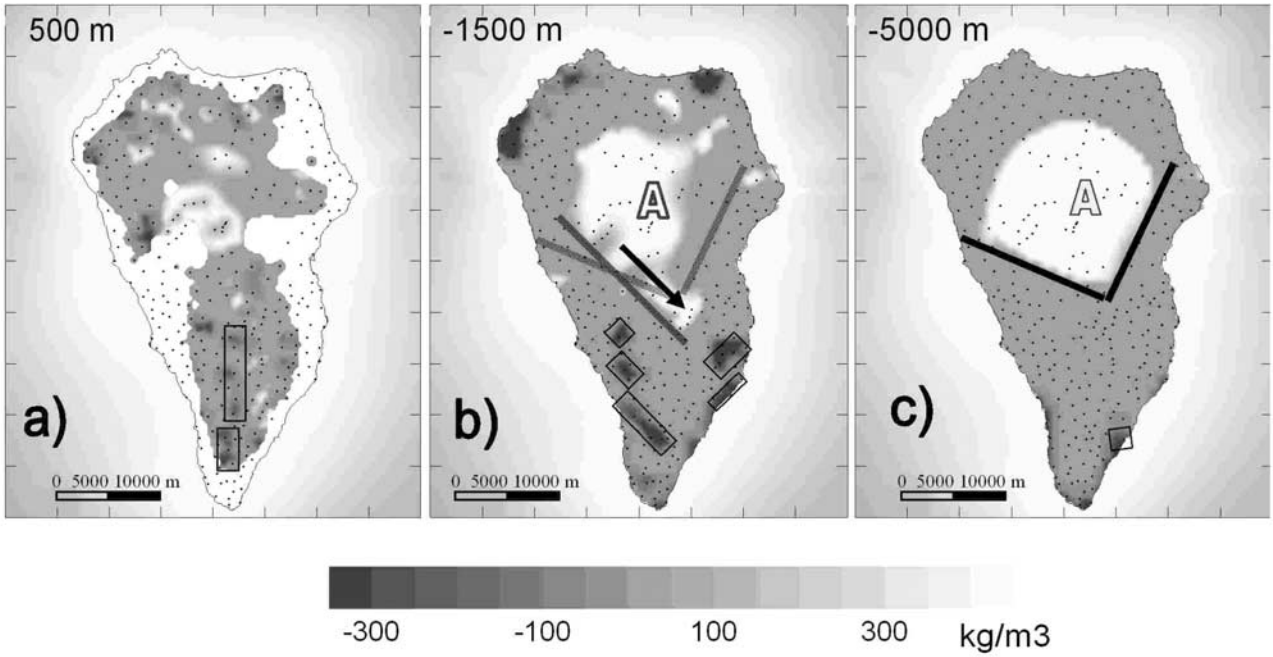

Figure 9. Some alignment and particular structures identified in the cross sections of the inversion model for height 500 m and depths 1500 m and 5000 m bsl. The high-density body "A" is identified for all levels. The low-density bodies located in the southern half show different features.

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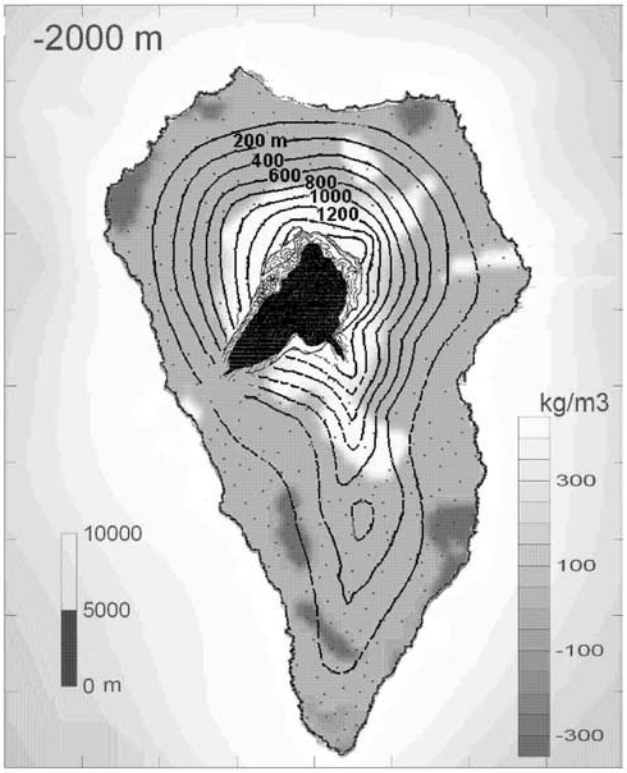

Figure 10. Overlapping of the lines for depth of the water table with the anomalous mass distribution corresponding to the inversion model for 2000 m bsl. A similar morphology is observed, suggesting the effect of lithologies as sustaining the subsoil water regime.

[2006] concluded that the old Taburiente/Cumbre Nueva and the young Cumbre Vieja volcanic systems are not only spatially separated but also represent two distinct volcanoes with separate magma plumbing systems.

[39] Worth pointing out is the good agreement obtained for the inversion model with respect to the groundwater table levels. The adjusted anomalous high-density structure constitutes the deep supporting basement which determines the shape and storage of the groundwater aquifers at La Palma.

[40] As regards the low-density areas, we associate the shallow (close to sea level) alignments of minima in Figures 9a and 9b with zones of shallow fractures associable to the N–S rift structure in Cumbre Vieja. Also, at a mean depth of about 2000 m bsl we detect some interesting minima below the western slope of Cumbre Vieja, not far from the zone where a large collapse is conjectured. They are arranged as en echelon planar tracks with azimuth N130°E, drawing a predominant strike course of N175°E. This morphology evidences a different dense material probably inherited from the coevent and postevent deposits of the Cumbre Nueva debris avalanche. A recent InSAR study using ERS-1/2 ESA satellite radar images detected some extended subsidence in this zone [*Fernández et al.*, 2008] suggesting an active dislocating process coincident with the shape and depth of this low-density body in the western side of Cumbre Vieja. Vertical profiles in Figures 8c and 8d show morphology of lateral slopes and fracturation areas in Cumbre Vieja. In the northern zone, more ancient and consolidated, these slopes are not observed, at least close to the shore.

[41] A strong minimum is also noticeable from the offshore (satellite) data for gravity disturbances southward close to the island. It may be related with a proposal for the location of a high-temperature/low-density anomaly in the upper mantle below Cumbre Vieja connected with recent volcanism, as also suggested by the long alignment of seamounts inferred from bathymetric data southward to La Palma [*Sandwell and Smith*, 1997]. Figure 8, corresponding to the inversion model, predicts a large external anomalous structure at a depth of about 10 km bsl This result agrees with *Galipp et al.* [2006], who, from analysis of fluid inclusions, suggest the existence of a short-time stagnation level at depths between 7 and 14 km close to Cumbre Vieja.

[42] With respect to the methodological aspects, here we have proposed some improvements with respect to a published inversion method. Results from the inversion can be interpreted in the framework of the geologic evolution of this ocean island volcano as a complex composite volcano and new unexpected features are enlightened upon, such as large thermal anomalies in the upper mantle southward of La Palma, as well as active fracture zones that could be related to the recent volcanism of Cumbre Vieja. The results obtained for La Palma as a test site show the interest of the methodology. A previous version of the former methodology has been applied to another volcanic islands (Canary Islands, Azores Islands) [*Camacho et al.*, 2001, 2007; *Gottsmann et al.*, 2008] and similar features (large intrusive core and zones of low density that can be associated with high thermal anomalies or less consolidated/fracturated zones) appear to be typical for this class of islands and can be adequately modeled by 3-D gravity inversion.

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